

Dr. Lukas Burk

RESEARCHER IN MACHINE LEARNING & BIOSTATISTICS

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Education

M. Sc. in Medical Biometry / Biostatistics

University of Bremen

Bremen, Germany

2021

- Thesis – A Comparison of Deep Learning Models for Energy Expenditure Prediction in Preschoolers

B. A. in Public Health

University of Bremen

Bremen, Germany

2018

- Thesis – Detection of Concomitant Use of Medications in GePaRD

Work Experience

Research Fellow

Leibniz Institute for Prevention Research and Epidemiology - BIPS

Bremen, Germany

Since 2026

PhD Student / Research Fellow

Leibniz Institute for Prevention Research and Epidemiology - BIPS / Chair of Statistical Learning and Data Science, LMU Munich

Bremen and Munich, Germany

2021 – 2026

Internship – Biometry

Leibniz Institute for Prevention Research and Epidemiology - BIPS

Bremen, Germany

2019

- Creation of a handout on the topic of count data and modeling approaches under zero-inflation and underdispersion
- Descriptive tables for validation purposes in the context of an ongoing study

Student Assistant

Competence Center for Clinical Trials (KKSB)

Bremen, Germany

2019

- Support during the analysis of study data
- Descriptive tables
- Exploratory linear regression modeling

Internship – Biometry, Epidemiology

Leibniz Institute for Prevention Research and Epidemiology - BIPS

Bremen, Germany

2016 – 2017

- Creation of an R / Shiny application for interactive data validation
- Data management and descriptive statistics

Teaching

Teaching assistant – Programming with Statistical Software (R)

LMU Munich

Munich, Germany

2022 – 2023

- Programming course aimed at students in statistics, data science, machine learning
- Preparation and assessment of programming exercises

Teaching Assistant – Quantitative Methods

University of Bremen

Bremen, Germany

2014 – 2018

- Tutorials for first year undergraduates in psychology
- Basics of descriptive and inferential statistics
- Application of these contents using R & RStudio
- Yearly course “Introduction to R”

Introduction to R

BIPS / RKI CARE Program

Bremen, Germany

2025

- Developed and held short course in cooperation with RKI CARE program
- Materials available on GitHub

Introduction to Machine Learning

BIPS, held by Marvin N. Wright

Bremen, Germany

2021, 2023, 2025

- Assisted with course as part of UBRA DataTrain program
- Course materials available on GitHub

Publications

Published Articles

- **L. Burk**, F. K. Meinzer, G. Casalicchio, M. N. Wright, and B. Bischl, “Xplainfi: Feature Importance and Statistical Inference for Machine Learning in R”, *The R Journal*, vol. 18, no. 3, p. 360–380, Sep. 2026, doi: 10.32614/RJ-2026-054.
- **L. Burk**, J. Zabolos, B. Bischl, A. Bender, M. N. Wright, and R. Sonabend, “A Large-Scale Neutral Comparison Study of Survival Models on Low-Dimensional Data”, *Bioinformatics*, p. btag186, Apr. 2026, doi: 10.1093/bioinformatics/btag186.
- S. Fischer, J. Zabolos, R. Sonabend, M. Becker, M. Lang, M. Binder, L. Schneider, **L. Burk**, P. Schratz, B. C. Jaeger, S. A. Lauer, L. A. Kapsner, M. Mücke, Z. Wang, D. Pulatov, K. Ganz, H. Funk, L. Harutyunyan, P. Camilleri, P. Kopper, A. Bender, B. Zhou, N. German, L. Koers, A. Nazarova, and B. Bischl, “Mlr3extralearners: Expanding the Mlr3 Ecosystem with Community-Driven Learner Integration”, *Journal of Open Source Software*, vol. 10, no. 115, p. 8331, Nov. 2025, doi: 10.21105/joss.08331.
- K. Blesch, N. Koenen, J. Kapar, P. Golchian, **L. Burk**, M. Loecher, and M. N. Wright, “Conditional Feature Importance with Generative Modeling Using Adversarial Random Forests”, *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 15, p. 15596–15604, Apr. 2025, doi: 10.1609/aaai.v39i15.33712.
- **L. Burk**, A. Bender, and M. N. Wright, “High-Dimensional Variable Selection With Competing Events Using Cooperative Penalized Regression”, *Biometrical Journal*, vol. 67, no. 1, p. e70036, 2025, doi: 10.1002/bimj.70036.
- H. J. Coyle-Asbil, **L. Burk**, M. Brandes, B. Brandes, C. Buck, M. N. Wright, and L. A. Vallis, “Energy Expenditure Prediction in Preschool Children: A Machine Learning Approach Using Accelerometry and External Validation”, *Physiological Measurement*, vol. 45, no. 9, p. 095015, Sep. 2024, doi: 10.1088/1361-6579/ad7ad2.
- M. Heckmann and **L. Burk**, “Gridsampler – A Simulation Tool to Determine the Required Sample Size for Repertory Grid Studies”, *Journal of Open Research Software*, vol. 5, no. 1, p. 2, Jan. 2017, doi: 10.5334/jors.150.

Preprints

- S. Fischer, **L. Burk**, C. Zhang, B. Bischl, and M. Binder, “Mlr3torch: A Deep Learning Framework in R Based on Mlr3 and Torch.” *arXiv*, 2026. doi: 10.48550/ARXIV.2604.18152.
- J. Piller, L. Orsini, S. Wiegrebe, J. Zabolos, **L. Burk**, S. H. Langbein, P. Studener, M. Goeswein, and A. Bender, “Reduction Techniques for Survival Analysis.” *arXiv*, Aug. 07, 2025. doi: 10.48550/arXiv.2508.05715.
- R. Sonabend, J. Zabolos, R. B. Bin, P. Kopper, **L. Burk**, and A. Bender, “Examining Marginal Properness in the External Validation of Survival Models with Squared and Logarithmic Losses.” *arXiv*, May 31, 2025. doi: 10.48550/arXiv.2212.05260.
- R. Sonabend, F. Pfisterer, A. Mishler, M. Schauer, **L. Burk**, S. Mukherjee, and S. Vollmer, “Flexible Group Fairness Metrics for Survival Analysis.” *arXiv*, Jul. 22, 2022. doi: 10.48550/arXiv.2206.03256.

Book Chapters

- G. Casalicchio and **L. Burk**, “Evaluation and Benchmarking” in *Applied Machine Learning Using mlr3 in R*, B. Bischl, R. Sonabend, L. Kotthoff, and M. Lang, Eds., 1st ed., CRC Press, 2024. doi: 10.1201/9781003402848.
- M. N. Wright, **L. Burk**, P. Golchian, J. Kapar, N. Koenen, and S. H. Langbein, “Machine Learning in Epidemiology” in *Handbook of Epidemiology*, W. Ahrens and I. Pigeot, Eds., Springer New York, pp. 1–68, 2024. doi: 10.1007/978-1-4614-6625-3_81-1.